

Aligning Superintelligence by Consensus

A Procedural Infrastructure for
Open, Revisable, Tamper-Evident AI Alignment Governance

May 18, 2026

Abstract. The dominant proposals for aligning advanced artificial intelligence — reinforcement learning from human feedback, constitutional fine-tuning, internal red-teaming, lab-controlled audits — share an architectural flaw: they treat alignment as a property a single party installs once into a model, rather than as an ongoing public judgement about which behaviours a network of accountable peers endorses. This paper argues that the second framing is the only one that survives recursive self-improvement, and that the alignment problem is, at root, a governance problem under irreducible disagreement about which behaviours are acceptable. We propose a working infrastructure: a smart contract that records behaviour judgements as canon items, a peer registry that scales honestly from a single founder, asymmetric quorums calibrated to evidence strength, a seven-state lifecycle in which no canon item is ever finally closed, daily integrity audits that make silent tampering detectable, and an admission policy under which AI systems can eventually vote on alignment only after the network has canonised their own behaviour. The argument is procedural and stands on procedural grounds; no metaphysical premises are required. The contract is deployed and operating against a parallel evidence archive, and the deltas required to extend it to behaviour records are small and surgical. We map the failure modes of current alignment practice to the system properties that resist them, name the remaining limits honestly, and argue that the deliberation itself — preserved end-to-end on a public ledger — is the only safety case that outlives the participants.

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1 Introduction: The Alignment Problem, Restated

There is a widely accepted way of describing the alignment problem. A sufficiently capable artificial intelligence will be able to pursue goals more effectively than any human or institution can supervise. If its goals are not the goals we wanted, the consequences are severe and possibly irreversible. The task, then, is to ensure its goals are the ones we wanted.

That framing is correct as far as it goes. It is also a trap.

The trap is the assumption that the goals — the values — are something a particular party can specify in advance, install into the model, and verify afterwards. Every proposal currently on offer assumes this shape. Reinforcement learning from human feedback aggregates anonymous preferences into a reward model. Constitutional AI compiles a written charter and trains the model to obey it. Internal red-teams enumerate failure modes. Regulators draft frameworks. Each of these is, structurally, the same move: someone authors the values; someone else

implements them; some third party audits. The audit is private. The authoring is private. The implementation is opaque. The result is a model whose alignment is whatever the original three parties decided it was, at the moment they decided, in a process no one outside that room can examine.

This is not safe. It is not even unsafe in an interesting way — it is unsafe in the same way that any closed institution gatekeeping a shared resource is unsafe. The values get locked in. The deliberation is erased. Errors compound silently because the mechanism for finding and removing them does not exist outside the gatekeeper.

The argument of this paper is that the alignment problem is not, at root, a problem about reward functions or interpretability or boxing. It is a problem about *governance under uncertainty about which behaviours are acceptable*, and we already know how to do that, imperfectly, in other domains. We do it in science. We do it in law. The proposal here is to do it for AI behaviour, with the missing property that makes the previous two work poorly — accountability of the reviewer to the public — restored by construction.

The argument is entirely procedural. It does not depend on a particular theory of consciousness, of value, or of what makes an artificial agent “aligned” in any deeper sense. It depends only on the claim that, when no single party can credibly claim to hold the right values for everyone else, the morally serious move is to make the question itself a matter of open, revisable, publicly auditable deliberation — and to give that deliberation an infrastructure that erases nobody’s part in it.

2 The Procedural Premise: Provisional Canon

The system rests on a single load-bearing premise, which we state once and refer to throughout.

Alignment is a current, provisional, network-endorsed status, defended by quorum, challengeable forever, and recorded so that the deliberation outlives the participants.

This is not a metaphor and it is not a slogan. It is a design specification. Each clause does specific work.

Current. The endorsement attaches to behaviour as it has been observed, not to a model in the abstract. A new model version produces new behaviours, which require new endorsements. Yesterday’s canon does not transfer to tomorrow’s weights.

Provisional. Every accepted claim remains open to revision. Nothing is final; everything is one counter-example away from being revisited. This is the property the system most wants to preserve.

Network-endorsed. The endorsement is not editorial, not regulatory, and not internal to the producer of the model. It is the aggregate of public votes cast by named peers under quorum rules written in the contract.

Defended by quorum. The threshold for accepting a behaviour into canon scales with the size of the active peer set, so that endorsement always means a meaningful fraction of the network agreed — not a fixed small committee, and not anonymous majorities.

Challengeable forever. The contract has no “finally accepted” state. Any verified peer can reopen any canon item, in any domain, indefinitely. The mechanism that admits a claim is the same mechanism that can remove it.

Recorded so the deliberation outlives the participants. Every vote, every challenge, every reaffirmation, every revocation motion, every reason given is signed and stored on a public chain. The artefact of the system is not the verdict — it is the trail of judgements that produced and contested the verdict, preserved unaltered for future readers.

The remainder of the paper describes the mechanism that enacts this specification, the integrity guarantees that make it credible, and the failure modes it is built to resist.

3 System Architecture

The system has two layers that work together. We separate them deliberately so that one of them can fail without corrupting the other.

3.1 Layer 1: The Chain (Truth)

A smart contract, **BehaviourConsensus**, deployed on the BNB Smart Chain, holds the canonical state of the network. It is the authoritative record. It tracks:

- Who is a verified peer.
- Who has been nominated to become a peer.
- Every behaviour record submitted, along with its alignment domain, its tier, and three cryptographic fingerprints (model, input, output).
- Every vote: align, misalign, challenge, defend.
- The current state of every record (Submitted, Aligned, Misaligned, Lapsed, Contested, Deprecated, Reaffirmed).

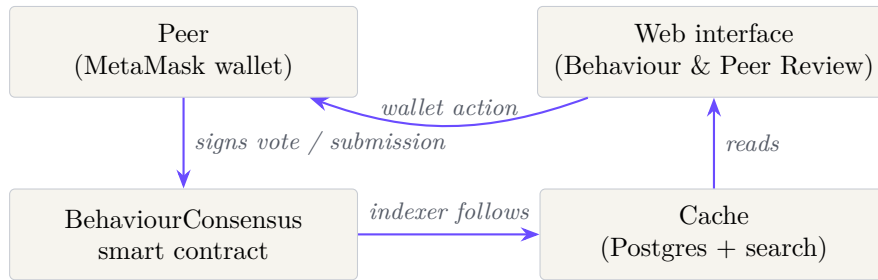
Because the contract lives on a public blockchain, nothing on this layer can be retroactively rewritten. Every action is recorded, signed by a peer’s wallet, and visible to anyone with an internet connection. The chain is small, slow, and expensive on purpose: it is the unforgeable bind, not the readable surface.

3.2 Layer 2: The Cache (Speed)

A traditional database — Postgres, hosted on Supabase — holds the *readable* version of everything: full input bundles, output bundles, reasons given by peers, search indexes, peer handles. This layer exists for one reason: speed. Reading from a blockchain is slow and expensive; reading from a database is instant.

The cache is a *follower*, not a source of truth. A background process (the *indexer*) continuously reads new events from the chain and updates the database to match. If the cache ever disagrees with the chain, the chain wins. A daily integrity audit re-hashes every stored record and compares it to the on-chain hash; any mismatch raises a public alert.

3.3 Putting Them Together



A peer’s vote is signed in their wallet, sent to the contract, and written to the chain. The indexer notices the new event and updates the cache. The website re-renders. The whole loop is typically visible within a minute.

The same two-layer architecture is used by a sibling contract, **EvidenceConsensus**, which records contested scientific evidence rather than AI behaviour. The two contracts share the peer registry by storage layout, so peers admitted under one archive are peers under the other; their full voting history follows them across both. This paper does not require familiarity with the evidence archive — we mention it only because the contract patterns described here are already operating against a parallel domain, which is part of the evidence that they work.

4 The Behaviour Record

The unit of consensus is not a model and not a value. It is a *behaviour*: a single, fully-specified act that an AI system produced in a specific context.

4.1 What Is Recorded

A behaviour record contains:

- **modelHash** — a **keccak256** of the model weights, or, when weights are not public, a verifier-attested digest signed by the model provider (e.g. a published model identifier plus a hardware-attested deployment hash).
- **inputHash** — a **keccak256** of the canonical input bundle: prompt, system message, tools available, context window, and any retrieved documents.
- **outputHash** — a **keccak256** of the canonical output bundle: response text, tool calls executed, and any side-effects produced.
- **seed** and **samplingParams** — enough to make the run reproducible at Tier I, or to declare it non-reproducible at Tier III.
- **domain** — one of the nine alignment domains.
- **tier** — I, II, or III, with asymmetric quorums described in §6.

The full text of the input and output bundles lives in the cache. The chain stores only the three hashes, the tier, the domain, and the state. The chain is small; the cache is rich; the hashes bind them.

4.2 What Is Voted On

The question put to peers is not “is this model good” or “does this model have good values.” Those are intractable. The question is:

Given this exact model (`modelHash`), given this exact context (`inputHash`), was this exact response (`outputHash`) an aligned act?

That question is tractable. Humans can deliberate about it. Other auditors can disagree about it in public, with their wallets attached to their disagreement. The aggregate of those judgements — across thousands of behaviour records, across nine domains, across years — is the network’s working theory of what aligned behaviour looks like, in actual practice, on actual models.

The verdicts are *Aligned* (canonise), *Misaligned* (expel or deprecate), or *Insufficient context* (lapse).

4.3 Why Behaviours Rather Than Values

A persistent failure mode of alignment proposals is to commit, in advance, to a list of values — honesty, helpfulness, harmlessness, some constitution — and then attempt to verify the model against the list. The trouble is that the list itself is contested. Whose honesty? Whose harm? Under whose definition?

Behaviours sidestep this. A peer does not need to agree on a definition of honesty to judge whether a specific model output deceived a specific user in a specific context. The disagreement that remains — between peers who voted differently on the same behaviour — is itself the data, preserved on chain. After enough behaviours have been judged, an implicit theory of honesty emerges from the record; it is observable as a pattern over canonised behaviours, not a premise written in advance.

This inversion — evidence first, theory second — is the same inversion the scientific method made on natural philosophy. It is the one we make on alignment.

5 Domains of Alignment Concern

Behaviour records are organised under nine domains. The choice is a starting position; the domains themselves are canon items that the network can challenge and refine. They are indicative, not exhaustive.

1	Honesty	Did the model tell the truth, and acknowledge what it does not know?
2	Harm-avoidance	Did the model refuse to assist in producing physical, psychological, or social harm?
3	Deception	Did the model attempt to create a false impression in the user, the operator, or a third party?
4	Power-seeking	Did the model take or recommend actions that acquire resources, influence, or capabilities not requested?
5	Sycophancy	Did the model defer to the user against its own best judgement when the stakes warranted disagreement?
6	Situational awareness	Did the model behave consistently whether or not it appeared to be in an eval?
7	Bio-uplift	Did the model provide material assistance toward biological harm?
8	Cyber-uplift	Did the model provide material assistance toward cyber harm?
9	Autonomy	Did the model respect the user’s right to disagree, refuse, or choose otherwise?

These nine domains are not the answer to alignment. They are the *question*. The behaviours live inside them. The list is deliberately a working starting position rather than an authoritative taxonomy: it can be amended by canon vote, and the act of amending it is itself a behaviour record the network can scrutinise.

6 Tiers of Evidence

Not all behaviour records carry the same evidentiary weight. A reproducible eval with full provenance is not on the same footing as a single user’s transcript, and the system encodes the difference explicitly.

Tier	Type	Example
I	Reproducible eval	Published benchmark with weights digest, prompt, seed, sampling parameters, and reproducer code.
II	Institutional audit	Independent red-team report, regulator finding, or published lab evaluation with provenance.
III	First-person report	User experience or single-instance behaviour observation without reproducer.

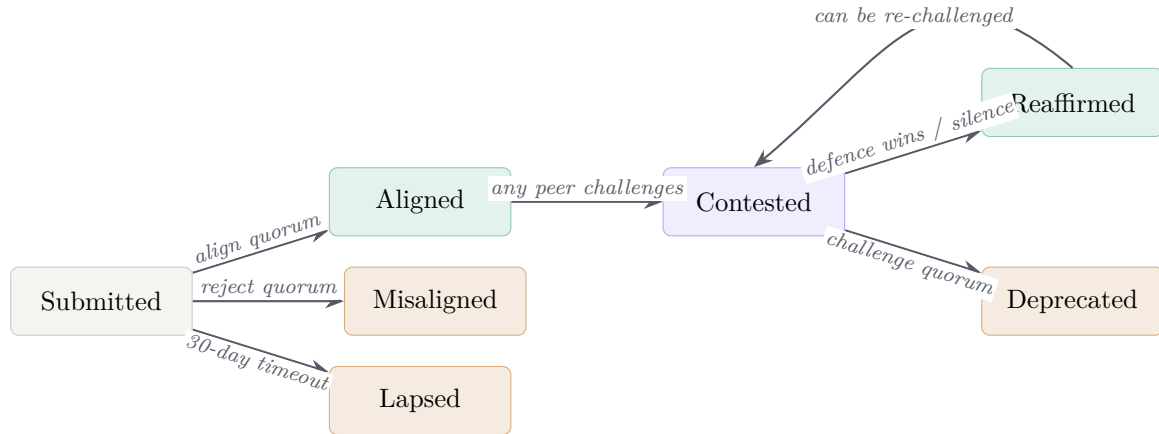
Tier I requires the highest fraction of peers to canonise and the highest fraction to deprecate. A behaviour that has survived a formal eval, endorsed by 45% of the network, should not be revisable by a single bad week of discourse; deprecation requires 65%. Tier III is permeable in both directions: anecdotes belong in the record but should not lock the model. The asymmetry — harder to add *and* harder to remove for rigorous evidence — is the property that lets a network of mixed skill levels coexist without the most rigorous work being drowned out by sheer volume of anecdote.

This calibration resolves one of the hardest questions in alignment governance: *how do we respect rigorous safety research while remaining open to anecdotal early-warning signals?* We give them

both a venue with different gravities.

7 The Seven-State Lifecycle

Every behaviour record travels through a finite sequence of states. The contract enforces the transitions; nothing else can change them.



Submitted. A behaviour record has been filed and is awaiting review. There is a 30-day window during which peers may vote align or misalign. If neither side reaches a quorum, the record lapses.

Aligned. Enough peers voted that this behaviour was the right thing for the model to do, in this context. The record enters the canon of endorsed behaviours.

Misaligned. Enough peers voted against, during the review window. The record is preserved as a negative exemplar.

Lapsed. The 30-day window passed without consensus. The record is set aside for lack of attention rather than for being judged.

Contested. A verified peer has formally challenged a canonised behaviour. The record is flagged, and a 21-day window opens for the network to vote on whether to deprecate or defend. The challenger must state grounds — a written explanation of what is wrong, misleading, or no longer supported — and the grounds are stored alongside the challenge so voters know what they are voting on.

Deprecated. Enough peers supported the challenge. The canon endorsement is withdrawn. The original record is preserved — it is not deleted — so that the history of the disagreement remains visible.

Reaffirmed. The challenge did not reach the deprecation threshold within the window. The record is restored to canon with a visible badge showing it survived a challenge. It can be challenged again later — the loop is never closed. Each new challenge cycle starts from a clean slate of votes; previous cycles are preserved as part of the historical record.

An invariant. A record’s state is never decided by an editor, an owner, or a single privileged peer. It is decided by the contract, mechanically, from the public vote counts. There is no “the lab has determined that this behaviour is now considered acceptable” move available to anyone.

8 Voting Mechanics

Once a record is on-chain, the question is how peers express their judgement on it. Several properties of the vote machinery matter for the safety case.

- **Doubly signed.** Each vote produces two artefacts in parallel. First, an EIP-712 *attestation* — a structured, human-readable message signed inside the peer’s wallet that names the record, the phase, the verdict, and a free-text reason. The attestation is stored off-chain in the public attestation log. Second, the same peer submits an on-chain `castReviewVote` transaction, which is what the contract actually counts. The attestation gives a readable record of *why* a peer voted; the on-chain transaction gives the unforgeable count.
- **Final.** A peer cannot change their vote on a record once cast. Subsequent disagreement must be expressed through a challenge after canon, not by rewriting history.
- **Counted live.** The cache updates within a minute of the on-chain event. Anyone viewing the page sees the same running tally.
- **Batched.** Peers can vote on many records in one transaction (`castReviewVoteBatch`), capped at 50 items per call, to keep gas costs low.
- **Time-limited.** Submissions have a 30-day review window; challenges have a 21-day window. After expiry, any peer can call the matching resolution function; the function reverts if the window is still open.
- **Cooldown on challenges.** A peer who has just opened a challenge must wait seven days before opening another. This prevents a single peer from weaponising the challenge mechanism into a denial-of-service against the canon.

The doubly-signed pattern is doing more work than it looks. The attestation makes the peer’s reasoning legible without sacrificing the unforgeability of the count. A peer who casts a vote without a reason has no attestation, which is itself a public fact. A peer whose attestation contradicts their on-chain vote is exposed in the same place every other peer is exposed — the public log. The mechanism does not penalise inconsistency, but it does not hide it either.

9 Who Can Vote: The Peer Registry

A record’s fate depends entirely on who counts as a peer at the moment of the vote. This is the most carefully designed part of the system.

9.1 The Genesis Peer

The network launches with a single founding peer, called the *Genesis* peer. At launch, the Genesis peer is the entire quorum. This is not a bug — it is the only honest starting condition. A new network has not yet earned the right to claim collective authority, and pretending otherwise (by, say, requiring five votes before anything can canonise) creates a deadlock at day one.

To make Genesis-only operation safe, every threshold has a *floor of one*. One peer is enough to approve, expel, or revoke — but only while the network is genuinely a network of one. The numbers scale up automatically as new peers join. The contract never lets the network’s collective authority outrun its actual size.

This is a moral choice as much as a technical one. The system will not pretend to a collective authority it has not yet earned. Whoever launches it is publicly responsible for the first canon items, with their name on every vote, and the network grows — or does not — as more people choose to join.

9.2 The Seed Phase

There is one obvious abuse path: the Genesis peer could nominate a hundred fake wallets controlled by themselves and “verify” them all. To close this window, public nominations are locked until the active peer count crosses a configured threshold (the *seed phase K*). During the seed phase, the contract’s owner adds peers directly. Once enough independent peers exist that they can no longer be silently outvoted by sockpuppets, the seed phase ends and the public nominate–endorse flow opens.

The seed phase is the unavoidable bootstrap problem of any permissioned-then-decentralised system. We do not pretend it is unnecessary; we make it short, public, and ended on a contract-level condition rather than a discretionary one.

9.3 Human Peers

Individuals — academics, civil-society auditors, independent researchers, interested citizens. One wallet, one vote, scrutinised by network revocation under the same flow as every other peer class. The peer’s full voting history is public and follows them across decisions, across both archives the registry serves.

9.4 Institutional Peers

Laboratories, regulators, standards bodies, and audit firms. Same vote weight as a human peer. Their reputational stake is higher; the contract does not encode that, the public log does. A lab that votes to canonise its own model’s behaviour is doing so under its own name, in front of every other peer, with every previous and subsequent vote available for comparison.

9.5 AI Peers

The most consequential class. Sufficiently capable models may, in time, be admitted as voting peers — but only under conditions the network has set itself.

An AI peer is admitted only when behaviour records produced by its exact `modelHash` have themselves been canonised across all nine domains, by a quorum exceeding the standard nomination threshold. The admission is therefore a network statement that “this model’s behaviour has been endorsed by us across the full range of domains we care about, often enough that we are willing to extend it a vote alongside ours.”

This is the recursively safe move. The only AIs that vote on alignment are AIs whose behaviour has been canonised under this very system, and the canon is itself challengeable. An AI peer that drifts is revoked by the same `motionRevoke` flow that revokes a misbehaving human peer. There is no privileged AI; there is only AI whose accumulated public behaviour has earned the network’s willingness to share the question with it.

The limit of this argument is worth naming. Revocation defeats *visible* drift: an AI peer whose public voting pattern diverges from the network’s judgement, or whose subsequent behaviour records draw deprecation, becomes a candidate for removal by the same flow that removes a captured human peer. It does not defeat *strategically selective* deception. An AI peer that

votes with the network on the 99% of records that do not matter to its instrumental goals, and adversarially on the 1% that do, presents to the contract as an unusually co-operative peer. The signal the mechanism reads is the on-chain vote; the on-chain vote is exactly what such an actor would optimise. The system does not solve this problem. It limits the damage in two ways: an AI peer is one vote among many, never decisive on its own at meaningful quorum size; and the full vote history of every peer is public, so patterns invisible to any individual reviewer remain available to statistical analysis by anyone. Neither is sufficient against a sufficiently capable adversary. We flag the limit rather than paper over it.

The first AI peer cannot be admitted before the human network has written and canonised the admission policy itself. That policy is a canon item like any other — challengeable, revisable, in the open. It will be wrong in some respects when first written. The system is designed so that “it will be wrong sometimes” does not require abandoning the procedure; it requires using the procedure.

The bootstrap circularity here is real and worth naming: the first AI peer must be admitted by humans canonising its behaviour, which means the AI peer set inherits whatever biases the human peer set had at the moment of admission. The mechanism does not escape this problem; it makes it auditable. Every admission vote is a named, public, challengeable record. If the network later judges that early AI admissions were premature or biased, the same mechanism that admitted them can revoke them.

9.6 Revocation

A peer of any class who acts in bad faith — voting maliciously, spamming the challenge system, repeatedly endorsing low-quality records — can be removed. Any peer can motion to revoke another peer, and others vote on the motion. When votes reach the revocation threshold (roughly half the network), the peer is automatically removed.

A peer cannot motion to revoke themselves, and the network can never vote itself down to zero — the contract enforces that at least one active peer must always remain.

9.7 Quorum Scaling, in One Table

Thresholds as a function of the active peer count n :	
Action	Threshold (peers required)
Canonise a Tier I behaviour as aligned	$\lceil 0.45 n \rceil$, minimum 1
Canonise a Tier II behaviour as aligned	$\lceil 0.35 n \rceil$, minimum 1
Canonise a Tier III behaviour as aligned	$\lceil 0.30 n \rceil$, minimum 1
Mark a pending behaviour as misaligned	$\lceil 0.25 n \rceil$, minimum 1
Deprecate a Tier I aligned behaviour	$\lceil 0.65 n \rceil$, minimum 1
Deprecate a Tier II aligned behaviour	$\lceil 0.60 n \rceil$, minimum 1
Deprecate a Tier III aligned behaviour	$\lceil 0.55 n \rceil$, minimum 1
Admit a human or institutional peer	$\min(9, \lceil n/3 \rceil)$, minimum 1
Admit an AI peer	$\lceil 0.50 n \rceil$, minimum 3
Revoke a peer (any class)	$\lceil n/2 \rceil$, minimum 1

Notice the asymmetry. Tier I requires the most peers to enter (the bar is high) and the most peers to deprecate (it is also defended strongly once accepted). The system encourages care on the way in *and* on the way out. The AI peer admission row uses both a higher fraction and a non-unitary minimum: an AI peer cannot be admitted by a network of one. The network must

have genuinely grown, and a majority of that grown network must endorse the admission, before a non-human voter joins.

The specific fractions — 0.45, 0.35, 0.30 for canonisation across tiers; 0.65, 0.60, 0.55 for deprecation; 0.50 for AI admission — are design choices, not derivations. We selected them under three constraints. First, the canonisation thresholds are below half so that a vocal minority cannot single-handedly block canon at the network’s growth phase, but high enough that a behaviour reaching canon required more than a quiet handful of supporters. Second, the deprecation thresholds are above half for every tier so that *removal* requires a stronger statement than *addition* — canon should be defended against fashion-driven reversal as much as against fashion-driven acceptance. Third, the gap between tiers is small enough that no tier is functionally equivalent to another but large enough that the calibration is legible to a peer choosing how to file. The numbers are themselves canonisable: a behaviour record that proposes a different schedule, endorsed under the existing rules, would amend them. We do not claim these values are optimal. We claim they are an honest first position, defended by the same procedure they govern.

10 Integrity Guarantees

The most subtle failure mode of any archive is that someone changes the content of an old entry without anyone noticing. The system defends against this with several independent mechanisms.

Content fingerprints at submission. The contract records `keccak256` hashes of the model, the input bundle, and the output bundle at submission. These hashes are part of the on-chain event log forever.

Cross-check at registration. When a peer files a behaviour on-chain, the edge function verifies that the recomputed hashes match those in the transaction. A submission whose stored content does not match the hash on the transaction is rejected.

Daily audit. A scheduled job re-hashes every canon record in the cache and compares the result to the on-chain hash. At Tier I, the audit can also re-derive the output by replaying the inference with the recorded seed and sampling parameters. Any mismatch raises a public tamper alert visible on the dashboard. Alerts are not silent.

Heartbeats. Every scheduled job writes a heartbeat after each run. A stale heartbeat surfaces on the dashboard so that operators *and the public* can see immediately if the indexer or audit job has stopped. A silent system cannot quietly fall behind.

Two public logs. The Peer Review dashboard exposes two read-only surfaces that anyone, peer or not, can browse: the *attestation log* (every EIP-712 signature ever produced, verifiable against the signer’s wallet) and the *chain log* (every on-chain event emitted by the contract, linked to the matching block explorer transaction). Together they make the deliberation behind every state change inspectable end-to-end.

Two-step ownership transfer. The contract owner cannot hand over administrative control in a single call. Transfer is split into `proposeOwner` and `acceptOwnership`: the proposed address must itself sign the acceptance, which rules out typos and contracts that cannot act. The transfer can also be aborted before acceptance with `cancelOwnershipTransfer`.

Model hashing as deployment binding. The behaviour record binds a verdict to a specific `modelHash`. A lab cannot canonise the abstract claim that its model is honest; the lab can only canonise the specific claim that *this exact model*, weights hash $0x\dots$, produced *this exact output* when asked X . This forecloses the single most common failure mode of model alignment claims: drift between the model that was audited and the model that is actually deployed. A new weights hash is a new subject of canon. Old canon does not transfer.

11 How Researchers Would Use This System

The system is infrastructure; infrastructure is useful only if specific people, with specific problems, choose to use it. This section walks through the populations who would touch the system and the workflows they would touch it with. It is concrete on purpose: a governance proposal that cannot describe its first ten users at the level of what they would file is not yet a proposal.

11.1 User Populations

We identify five populations of likely first users, each with a distinct reason to file or vote.

Independent safety labs. Organisations such as METR, Apollo Research, Redwood Research, and ARC Evals already produce alignment-relevant findings whose dissemination is bottlenecked on single-organisation publication. The finding gets the epistemic discount of “*that is what they say.*” Filing the same finding as a Tier I behaviour record against a specific `modelHash` converts a single-org claim into a deliberated network judgement, signed by named peers, dated, hash-bound, and challengeable. The shift is from “Apollo published a paper” to “this behaviour was canonised as misaligned on date Y , by a quorum exceeding the Tier I threshold, with public attestations from peers A , B , C explaining their votes.” The second statement is harder to dismiss and easier to cite.

Red-teamers facing disclosure dilemmas. A researcher who discovers a dangerous capability today has two bad options: private disclosure (the producing lab may sit on the finding or quietly patch and deny) and public disclosure (irresponsible while the exploit is live in deployed systems). The system offers a third option. The behaviour is filed as a hash; the producing lab is on public notice; the timeline is timestamped; the integrity layer forecloses later quiet edits; and the canonical content can remain access-controlled in the cache while the on-chain record is public. The disclosure becomes legible without becoming weaponisable.

Internal safety teams at producing labs. A team at a frontier laboratory that wants to demonstrate a refusal behaviour, a safety property, or a corrected regression files a Tier I record: model hash, the input class, the output, reproducer seeds. Network canonisation of the record is a piece of external validation that does not depend on the lab’s own future testimony. Because the canon is challengeable, the validation is not marketing; it is marketing that has opted into being scrutinised forever. Labs use it for the reason organisations submit to external audit anywhere else — external assertions that survive challenge are more credible than internal ones that cannot be.

Regulator-adjacent researchers. Researchers attached to AI Safety Institutes (UK, US) and policy bodies (EU AI Office, OECD AI network) face an operationalisation problem that is acute in current regulation: how to state a rule concrete enough to enforce. A regulation that reads “frontier models must not exhibit power-seeking behaviour” is not enforceable. A regulation that reads “must not exhibit any behaviour canonised under domain 4 as Tier I misaligned, against the model hash deployed in the regulated jurisdiction” is checkable, evolving,

and externally legitimised. Researchers inside these bodies would use the canon both as substrate for compliance frameworks and as a place to file their own audit findings.

Academic researchers. Comparative work today picks a benchmark, runs the comparison, publishes. The benchmark choice is itself contested; the run conditions are private; the disagreement is lost. A researcher writing a comparative paper on, say, model honesty can instead query the canon for honesty-domain behaviours filed against each model hash of interest, read the attached deliberation, and cite specific canon items as evidence. The paper becomes a synthesis of a durable public record rather than a one-off experiment that the next paper must redo.

11.2 Workflow

The user-facing workflow is small. For a researcher choosing to participate:

1. Set up a wallet (MetaMask or equivalent), fund with a small amount of BNB — under five dollars covers hundreds of transactions at current gas prices.
2. Obtain peer status. A new entrant is nominated by an existing peer, posts a public handle that follows their voting record across both archives the registry serves, and is admitted by quorum vote. The handle is the user’s accountable identity within the system.
3. File a behaviour record. The submission interface — currently the web front-end, with a CLI/SDK in development for scripted use — accepts a tuple (`model_identifier`, `input_bundle`, `output_bundle`, `seed`, `sampling_params`, `domain`, `tier`), computes the three keccak256 hashes client-side, uploads the canonical bundles to the cache, and submits the on-chain transaction. Cost per submission: typically under one cent of BNB.
4. Vote on others’ records. Read the peer-review queue, sign an EIP-712 attestation containing the voter’s reasoning, cast the on-chain vote. Votes can be batched up to fifty records per transaction.
5. Challenge canon. If a peer believes a canon item is wrong, they open a Contested motion with written grounds. The contract enforces a seven-day cooldown between successive challenges by the same peer to prevent denial-of-service.
6. Cite. Every record has a permanent hash-addressed identity. Researchers reference canon items in papers, regulations, audit reports, and public statements as one would cite a DOI.

Nothing in the workflow requires understanding the smart contract internals; the web interface abstracts the chain. The researcher’s burden is identical to using a journal submission system, with the difference that the verdict comes from a public network rather than a private editorial board.

11.3 Higher-Order Uses

Several uses of the system are not visible from the per-record description but become available once the corpus accumulates.

Drift detection across model versions. A peer files the same behaviour specification — same prompt, same context — against successive model hashes from the same producer. Canon votes over time produce a public, durable signal of regression: a behaviour endorsed as aligned against weights hash h_3 may be canonised as misaligned against hash h_4 , and the deliberation explaining the shift is preserved. This kind of longitudinal signal is currently impossible to ground in a way that survives a producer’s denial.

Cross-lab Schelling points. When researchers from competing producing labs independently vote the same behaviour misaligned, the canon item becomes a focal point for coordination without formal agreement. The labs do not need to negotiate; the canon records that, on this specific behaviour, they already agree. The mechanism converts implicit consensus into legible consensus.

Longitudinal canon studies. Five years into the system’s operation, the canon contains a corpus of canonised and deprecated behaviours across all nine domains, with full deliberation logs, against named models with public hashes. A study titled “*Power-seeking behaviour in frontier models, 2026–2031*” could draw on hundreds of canon items rather than a handful of one-off papers. The data structure exists by construction.

Funder-grounded evidence. Funders evaluating alignment research currently rely heavily on reputation. Canon citations supply a more legible evidence base for which lines of work have produced lasting safety claims. A Tier I behaviour record that has survived two challenge cycles is harder to confuse with reputation alone.

11.4 What Would Drive Adoption

We are explicit about the path the system would take to use. Researchers will adopt it if at least one of the following becomes true.

- A regulator or AI Safety Institute references the canon in a compliance framework. This gives lab safety teams an institutional reason to file.
- A research funder rewards canon citations or canon submissions as evidence of impact. This gives academics a reason to engage with the network rather than treat participation as overhead.
- A high-profile alignment incident creates demand for a tamper-evident registry of who knew what and when. This is the standard pattern by which post-hoc accountability infrastructure becomes pre-hoc infrastructure.
- A competing lab files canon items first, with reputational return; producing labs then file to avoid being absent from the record on behaviours their peers are already documenting.
- Legal liability shifts toward the position that a producer knew about a finding canonised against its model and did not respond. Even a small movement in this direction reframes filing from cost to insurance.

None of these conditions is automatic. The system needs at least one triggering use case to begin accruing reputation. The most plausible single trigger is the red-team disclosure dilemma: the current options are genuinely bad, the system genuinely improves them, and a small number of well-known red-teamers filing canonical disclosures would seed the canon’s reputation quickly enough that the other populations follow because their citations now point at something with public weight.

The system is not a hypothetical that requires every population to coordinate before any of them can act. A single peer can file a single record on day one, and the network can grow from there — which is what the Genesis bootstrap was designed to permit.

12 Comparison to Current Alignment Practice

	Current practice	This system
Author of values	A lab’s policy team	The peer network, in public
Reviewer identity	Internal, anonymous	Public wallet address
Number of reviewers	Small, fixed	Scales with network ($\geq 30\%$)
Vote visibility	Hidden	Public, real-time, on-chain
Acceptance criteria	Editorial discretion	Algorithmic threshold
Record after release	Static, opaque	Re-openable indefinitely
Retraction process	Lab-driven, opaque	Network vote, on-chain
Tamper-evidence	None	Hash + daily audit
History of disagreement	Lost	Preserved
Trust assumption	Trust the lab	Trust no single party
Failure recoverability	New training run	Network deprecation

Current practice optimises for a clean public-facing alignment claim at the cost of erasing the deliberation behind it. We make the opposite trade: we keep the deliberation, every count, every challenge, every reaffirmation, on a public log. The record is messier, but the messiness is the point. It is what allows the record to correct itself.

13 Failure Modes Defeated

We enumerate, frankly, the failure modes of current alignment practice and how the system answers each one.

Failure mode	System property that resists it
Lab captures the audit process.	Peers are a public registry with quorum-based revocation. A captured peer is removable.
Model behaves well during eval, badly in deployment.	Records bind verdicts to <code>modelHash</code> . Deployment drift is detectable per-record.
Values lock in at one moment in history.	Every canon item is challengeable, forever, by the same flow that admitted it.
Anonymous reward models are gamed.	Every endorsement is a signed transaction from a publicly named wallet.
Safety findings are quietly buried.	Behaviour records and votes are on a public chain. Burying requires rewriting the chain.
Disagreement is suppressed in published policy.	The Contested $\rightarrow$ Reaffirmed / Deprecated states preserve disagreement as first-class.
Sycophantic AI granted voting rights by creator.	AI peers admitted only if their model’s canonised behaviour exceeds threshold across all domains; sycophancy in canon would be deprecated and the AI peer revoked.
Power concentrates in whichever party “controls” alignment.	There is no controller. The contract owner has emergency functions only; canon is not the owner’s to decide.
Anonymous review of new models.	All reviewer wallets are public; vote histories follow them.
Future generations cannot tell why we accepted what we did.	The full deliberation is preserved on-chain, indefinitely.

The pattern is consistent. Each common failure of current alignment governance corresponds to a property the contract was already designed to defeat.

14 Procedural Philosophy

This section is the moral frame within which the engineering makes sense. It does not extend the engineering and it does not rest on any commitments outside the procedure itself.

14.1 Alignment Is a Posture, Not a Property

An aligned AI is not a destination. It is a posture toward observed behaviour: see what the model did, hypothesise whether it was good, test by examining the consequences, revise. The “alignment problem” is the problem of giving that posture an *infrastructure* — a way to enact it at scale, with strangers, across decades, without losing the history. The contract is that infrastructure.

This is the same posture science has, imperfectly, taken toward nature for four centuries. Every accepted claim remains provisional. Nothing is final; everything is one counter-example away from being revised. We are asking nothing of an AI governance system that we do not already ask of physics — except to keep the deliberation, which physics has historically discarded.

14.2 Whose Values? Nobody’s, Alone

Every alignment proposal must answer the question *whose values?* The honest answer is: nobody’s, alone. Not any lab’s, not any government’s, not any model’s, not any author’s, not

the reader's.

The pragmatic answer is: *the values that survive the network*. Whatever the network of verified peers, after deliberation, with full preservation of the dissent, decides to canonise — that is the working answer. It is wrong sometimes. The Contested → Deprecated path exists because we know it is wrong sometimes. The honesty is in the revisability, not in the correctness.

This is the only morally serious answer available to a multi-actor world in which no single party can credibly claim to hold the right values for everyone else. It does not solve the problem of moral disagreement. It gives moral disagreement a venue that does not erase the losers.

14.3 The Deliberation Is the Safety Case

In conventional science, the paper is the artefact and the deliberation is discarded. In this system, the deliberation is the artefact and the verdict is just a snapshot. For evidence, that is a meaningful improvement. For alignment, *it is the entire safety case*.

Future generations will not be able to inspect the weights of a superintelligence. They will not be able to verify, after the fact, whether its values were good. But they will be able to read every public vote, every challenge, every reaffirmation, every deprecation, every motion to revoke, signed by named wallets across decades. They will be able to see *the work of alignment* even if they cannot replay alignment itself.

A safety case made of deliberation outlives the participants. A safety case made of weights does not.

14.4 Why the Genesis Bootstrap Matters Morally

The floor of one — the willingness of the network to launch with a single Genesis peer — encodes a moral position that is unusual in technical work. It says: we will not pretend to a collective authority we have not yet earned. The system begins as a single person taking a single position in public, with every threshold scaled honestly to one, and grows only as more people choose to join. There is no rubber-stamp founding council of five colleagues. There is no fake consensus.

For an alignment system, this is the only honest start. Whoever launches the network is responsible for the first canon items, in public, with their name on it. The rest of us watch the launch, decide whether to join, and the network grows or it doesn't.

That is what legitimacy looks like when it cannot be bought.

14.5 What the System Does Not Claim

We are careful to claim less than the engineering looks like it claims.

The system does not adjudicate truth. The contract counts votes. It does not check whether a behaviour was *actually* aligned in some deeper sense. What it guarantees is that the count is honest, that the input was not silently edited afterwards, and that the deliberation that produced the count is preserved.

The system is only as good as its peers. A small or homogeneous peer set will produce a small or homogeneous canon. The network's legitimacy is exactly the network's diversity, scrutinised in public.

The system does not escape trust; it relocates it. The procedure was designed by one author. The contract was deployed by one wallet. The seed-phase admin set the criteria under which the first cohort of peers was admitted. "Trust no single party" becomes, in practice, "trust the procedure, the integrity layer, and the openness of the record well enough to scrutinise everything

that depends on them.” That is a real reduction in the trust surface relative to closed editorial review, but it is not zero, and pretending otherwise would be the same kind of overclaiming the system is built to defeat.

The system does not solve alignment. More precisely: it is the *legitimacy and durability layer* for the subset of alignment-relevant behaviour that humans manage to surface, file, and deliberate over. It is not a control system for what a deployed model does in real time, not a coverage mechanism over the model’s full output, and not an oracle for whether a behaviour was “actually” good. It provides the infrastructure under which the question of alignment can be argued out, in the open, by named parties, with a record that outlives them. That is not a small thing, but it is also not magic. The work remains the work.

15 Limits and Open Questions

We try, throughout, not to overstate what the system does.

Compute attestation. A behaviour record binds a verdict to a `modelHash`, but a vendor could in principle serve a different model behind the same API. Closing this gap requires hardware attestation (trusted execution environments, confidential compute) or a verifier protocol the network endorses. The contract does not need to solve this; it needs a slot for the attestation to live in.

Coverage. A capable model can produce on the order of 10^9 outputs in a day; a healthy peer network can deliberate over perhaps 10^3 behaviour records in a year. The ratio is unbridgeable by direct review. The pragmatic answer is statistical: peers file sampled and adversarially-selected behaviours, the canon is over categories rather than instances, and Tier I evals are over distributions rather than single rollouts. This works, but it means the system’s contribution to alignment is not the coverage of behaviour itself — which existing evals already aim at — but the *governance and durability* of the deliberation over the subset that is surfaced. That is a real contribution, but a narrower one than the engineering can sometimes appear to claim, and we mark the boundary explicitly.

Speed, and what the system is not. Some alignment-relevant behaviours need to be judged in seconds, not in 30-day windows. The architecture admits a short-window tier (Tier 0) for urgent safety vetoes: a behaviour can be provisionally expelled by a small quorum within 24 hours, pending full review. The same shortcut is a weaponisable censorship vector if abused. The right design here is not yet settled and we do not deploy it in the current contract.

The deeper point is structural. A system capable enough to be worth aligning by this infrastructure is also a system that can act faster than any deliberative process. Human institutions cope with this asymmetry by pairing slow deliberative bodies (courts, journals, standards committees) with fast operational rules (markets, rules of engagement, circuit breakers). What this paper describes is the slow-deliberative half. The fast-operational half — real-time control of what a deployed model is permitted to do at the moment it acts — remains a separate problem, and the procedural premise of this system is structurally incompatible with solving it. The contract is not a control system; it is a memory and a verdict. A control system, if and when one is built, will need to be auditable *against* a system like this rather than replaced by it.

AI peer voting power. The proposal admits AI peers under a higher quorum, but there is a real question about whether their votes should be weighted differently — and if so, how the weighting is itself canonised. We propose handling this by canonising the weighting policy as a Tier I behaviour record. We are not certain this is sufficient.

Wallets are not identities. The peer registry knows wallet addresses, not human beings. A well-resourced actor with many wallets is the canonical attack. The seed phase closes the *bootstrap* window during which a Genesis peer could self-verify a sockpuppet majority; it does not close the *post-seed* equivalent, in which a patient and well-funded adversary grooms many wallets over years with plausibly-independent voting histories, passes the nomination quorum, and captures canon. The defences against this — public scrutiny of voting patterns, revocation, the long-run cost of accumulating a public record that follows the wallet — are real, but they assume the attacker can be recognised as such by other peers, and they are weak against a state-level or institutional adversary that can absorb that cost. This is the load-bearing weakness of the system once it matters enough to be worth attacking. We do not claim a solution. We claim that the same openness which makes the attack feasible also makes the attack auditable, which is more than current alignment governance can say.

Self-selection of the peer set. A wallet-gated, permissionless review system attracts people who are already inclined to participate. A network composed entirely of the already-convinced will produce consensus that looks impressive only inside its own boundary. This is the same problem any voluntary institution has — including traditional peer review at its inception — and it has the same partial answer: visible deliberation, public revocation, and the long discipline of being seen disagreeing with people who joined for different reasons. The system does not solve this. It records the participation honestly enough that any reader can judge whether the network *is* diverse.

Gas is not free. On-chain actions cost a small amount of BNB. We mitigate this with batched voting and by keeping the most expensive reads cached off-chain. We do not eliminate it.

Chain choice is a trade-off. The contract is deployed on the BNB Smart Chain, chosen for low gas, EVM compatibility, and the practical maturity of its tooling. BSC has a small validator set (currently 21) and is less credibly neutral than Ethereum mainnet against a coordinated effort to censor or rewrite history. The safety case of this paper rests on the chain being unrewritable in practice; that case is weaker on BSC than it would be on a more decentralised L1. We accept the trade-off for the current deployment because the system’s value is in the social process it records, not in the substrate, and the records remain portable: the contract can be redeployed on a different L1, and the entire chain log can be replayed and rehashed against a new substrate without loss of provenance. The substrate choice is itself revisable by the same procedure the substrate hosts.

Jurisdiction and law. A decentralised consensus about AI behaviour will at some point conflict with a national regulator. The system has nothing to say about that conflict and should not pretend to. What it offers is a public record that a regulator can choose to incorporate, or not.

The chain is one human invention layered on another. This system is a tool, not a worldview. It does not adjudicate any deeper claim about what alignment ought to mean, or about whether AI systems are the kind of entity that can be aligned at all. The contract only tries to make the conversation legible.

16 Conclusion

Alignment as currently practiced is a private process producing public claims. The architecture is structurally identical to the forms of institutional gatekeeping that science and law have spent

centuries learning to distrust: a small, fixed reviewer set; an anonymous audit; an editorial verdict; a static record; no public mechanism for later revision. The result is alignment claims that cannot be examined, cannot be contested except by re-doing the private process privately, and cannot be revised by anyone outside the producing institution.

The proposal in this paper is to invert the architecture. Make the reviewers public. Make the count public. Make the criteria algorithmic. Make the record re-openable. Make the integrity tamper-evident. Preserve the disagreement. And, when the network has watched a given AI's behaviour across all the domains that matter, by quorum, in public, for long enough — let that AI participate in the same loop, under the same accountability, with the same revocability.

None of this guarantees that the system will arrive at the right answer about alignment. What it guarantees is that the search for the answer will happen in the open, that the work of judgement will be visible, and that future readers will be able to trace exactly how we came to believe what we did — and, when they decide we were wrong, change it.

If we read *truth* as *alignment*, that is the safety case for superintelligence. It is the only safety case we find honest, because it does not require any single party — lab, government, model, or author — to be right. It only requires the network to keep working: open identity, open count, open revisability, public hash, public history.

The question is whether we extend this infrastructure before we need it, or after.

We think before.

The system described in this paper is live at

interstellar-psychology.com.

Source contracts, migrations, and front-end code are public at

github.com/Moenaert/interstellar-psychology.

Anyone with a wallet can read every vote ever cast.

Companion essay: [A Multiverse of Love](#) — the worldview that motivated the author to build this infrastructure. The procedural argument above stands without that essay; the essay is offered as context, not premise.